9903979160 • Mumbai, India
soumil.konar2001@gmail.com
GitHub • LinkedIn • Portfolio

SOUMIL KONAR

GENERATIVE AI ENGINEER

SKILLS

AI / ML

LangGraph • Semantic Kernel • RAG • Agentic AI • Prompt Engineering • LlamaIndex • NLQ-to-SQL • ONNX Runtime

LLMS & APIS

OpenAI API • BCAI API • ChromaDB • pgvector • MCP • FastAPI • Flask

LANGUAGES

Python • C#/.NET • TypeScript • JavaScript • SQL • Cypher • Rust • Bash

FRAMEWORKS & TOOLS

Node.js • React.js • Pandas • NumPy • sentence-transformers • PyTorch • TensorFlow

DEVOPS & CI/CD

GitLab CI/CD • Cloud Foundry / TAS • Docker • CredHub • REST APIs

DATABASES

PostgreSQL • SQL Server • Neo4j • ChromaDB • Oracle

VISUALIZATION

Tableau • Blender • AutoCAD

EDUCATION

JADAVPUR UNIVERSITY

M.Tech – Information Technology

Sep 2024 – Apr 2025
Kolkata, India

MAKAUT

B.Tech – Information Technology

Nov 2020 – Jun 2024
CGPA: **9.3 / 10**
Kalyani, India

SOUTH POINT HIGH SCHOOL

Class 12 (CBSE): **91%** • Class 10 (CBSE): **94%**

CERTIFICATIONS

- GATE 2024 – **Qualified** | CS&IT | Score: 43
- Software Engineering Program — **Goldman Sachs**
- DSA using Python — **NPTEL**
- Tools for Data Science — **Coursera**

PROFESSIONAL SUMMARY

Generative AI Engineer with **1+ year** of experience at TCS building production-grade LLM systems for Boeing's enterprise aviation platforms. Built **8 RAG microservices** using LangGraph, Semantic Kernel, ChromaDB, and Neo4j — including a plug-and-play multi-use-case orchestrator, graph ETL pipelines, and LLM document generation. Combines deep AI pipeline expertise with DevOps fluency (GitLab CI/CD, Cloud Foundry, CredHub) to ship scalable enterprise AI end-to-end.

PROFESSIONAL EXPERIENCE

TCS — *Mumbai, India*
AI Engineer — (April 2025 – Present)

- Built **8 production RAG microservices** (FastAPI + LangGraph/Semantic Kernel/ChromaDB) across 5 enterprise domains, deployed on Cloud Foundry/TAS with CredHub secret management.
- Architected **MDE-Rag** — descriptor-driven RAG orchestrator for an enterprise **schedule management system** with **classifier → plugin → 4-node LangGraph pipeline** routing across **5 use cases**; enables project managers to query schedule impacts, resource conflicts, and recommendations in natural language across **10,000+ task schedules**.
- Designed a **Neo4j graph ETL pipeline** (SQL Server → **10+ node-label graph**) with batch ingestion and predecessor parsing, enabling downstream schedule impact analysis.
- Engineered a **7-step LLM doc generation service** using pgvector semantic search, BCAI Batch API relevance scoring, and dual-level prompt chaining to produce ATA-formatted scope-of-work documents.
- Owned **GitLab CI/CD pipelines**, CredHub secrets, and Cloud Foundry deployments; built MCP servers exposing Neo4j and SQL Server as AI-queryable tools.

C-DAC — *Kolkata, WB*
Software Developer Intern — (Aug 2023 – Nov 2023)

- Built an automated audit report system delivering Stage 1 & 2 reports.
- Cut auditor workload by **50%** and halved report turnaround time by eliminating manual data compilation.

PROJECTS

MULTI-USE-CASE RAG ORCHESTRATOR — *Python, LangGraph, Neo4j, SQL Server*

- **Plug-and-play engine**: use cases defined by JSON descriptors (input, variations, data access, schema mapping); LLM plans and executes SQL, Cypher, and Python resolvers dynamically at runtime.
- Classifier routes across 5 use cases (schedule impact, resource recommendation, conflict detection, copilot, fallback) with confidence scoring; plugin registry auto-loads configs from YAML.
- Multi-source execution node dispatches SQL Server templates, Neo4j graph traversals, Excel lookups, and Python functions from a single unified execution step.

GRAPH ETL PIPELINE & MCP SERVERS — *Python, Neo4j, SQL Server, MCP*

- ETL script transforms relational schedules into a graph (**10+ labels, 8 relationship types**) with 500-node batch commits.
- Dual MCP servers let AI assistants query Neo4j (Cypher) and SQL Server (T-SQL) directly as tool interfaces.

ENTERPRISE RAG MICROSERVICES SUITE — *FastAPI, LangGraph, Semantic Kernel*

- **5 domain RAG services**: NLQ-to-SQL pipelines (DQN, AWCert, PMDB with LangGraph), multi-system routing across DQN and NCAT databases (CMM with Semantic Kernel tool-calling), and hybrid vector/metadata search (Flam with ChromaDB).
- Session tracking, follow-up detection (**6+ operation types**: filter, aggregate, paginate, keyword search), and standalone query rewriting for multi-turn context.

OUTFIT SUGGESTION WEBSITE — *Python, TensorFlow, JavaScript*

- Fashion recommendation engine combining NLP and image recognition for personalized outfit suggestions with LLM-based trend analysis and monthly model retraining.
- Built a web interface for interactive style exploration with context-aware suggestions adapting to user preferences and seasonal trends.